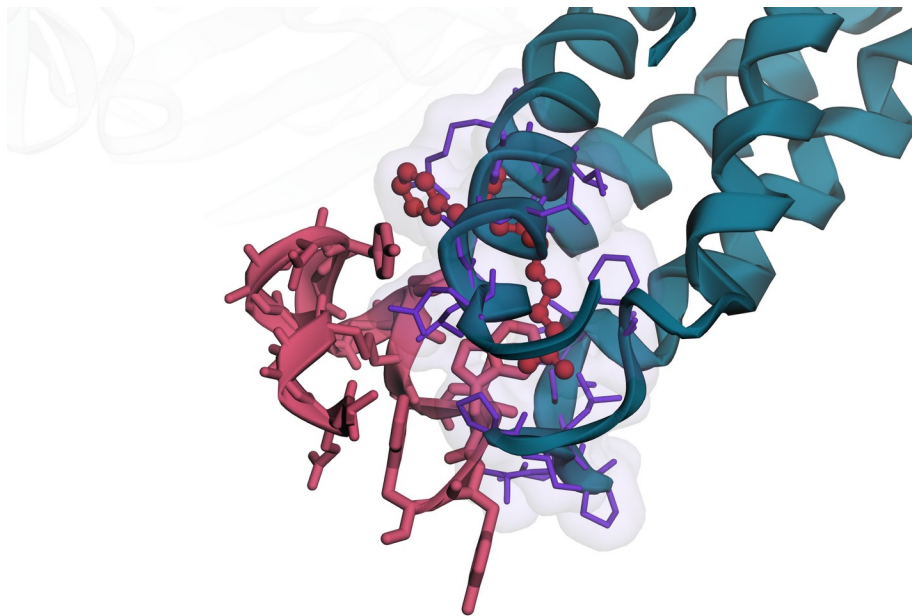


IL23R–IL-23 p19

small-molecule inhibition

Psoriasis target validation, structural strategy, discovery plan, and target product profile



PORTFOLIO RECOMMENDATION CONDITIONAL GO — run a gated conventional small-molecule feasibility campaign centered on the 8UUI inducible p19 pocket; retain an IL23R-facing mimetic track only as a higher-molecular-weight backup.

Research cutoff

1 September 2026

Evidence basis

Peer-reviewed primary literature, experimental PDB structures, FDA labels/reviews, and clearly flagged sponsor disclosures

Scope

Direct extracellular disruption of the p19–IL23R interaction; TYK2 inhibition is treated only as a comparator

Report map

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- 1 [1. Target biology and validation](#)
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How to read the report Observed evidence and proposed program thresholds are kept separate. All TPP values and stage gates are decision targets, not claims about an existing molecule. Cross-trial efficacy comparisons are descriptive only.

Executive decision

Decision: conditional GO. Human disease biology, selective p19 blockade, and direct oral IL23R antagonism are strongly validated. The remaining question is narrower but difficult: can a non-peptide, orally practical molecule bind the extracellular complex with enough potency, selectivity, and exposure to compete with approved icotrokinra? PDB 8UUI changes this from a speculative flat-PPI project into a testable pocket-ligandability program, but it does not yet establish conventional small-molecule tractability. [6,10,13–15,29]

Table 1. Integrated target-assessment scorecard

Dimension	Assessment	Rationale
Disease / pathway validation	5.0 / 5	Human lesions, protective genetics, three approved p19 antibodies, and direct oral receptor blockade converge.
Human pharmacologic validation	5.0 / 5	Icotrokinra is FDA-approved and outperformed deucravacitinib on week-16 PASI 90 in two head-to-head trials.
Mechanistic / structural validation	4.5 / 5	Multiple receptor complexes define the interface; 8UUI provides an inducible p19 pocket and screening probe.
Conventional small-molecule tractability	2.5 / 5	The native interface is broad and shallow; public fragments bind only at ~0.18–1.95 mM and lack co-structures.
Assay and translation readiness	4.0 / 5	Direct binding, displacement, human whole-blood pSTAT3, skin challenge, and lesion signatures form a coherent cascade.
Differentiation headroom	2.5 / 5	Another oral product must beat icotrokinra on at least two of dose, food effect, efficacy, safety, cost, or convenience.

INVESTMENT THESIS Fund a gated feasibility campaign, not an open-ended PPI program. The lead route is fragment growth/linking into the cyclic-peptide-induced p19 pocket; functional p19-IL23R disruption, rather than pocket binding alone, is the first decisive gate.

What would create a differentiated medicine?

- A non-peptide tablet at ≤200 mg once daily as the minimum, with ≤50–100 mg and no mandatory fasting as the target.
- Human whole-blood IL-23 inhibition that is coverable through the dosing interval while preserving IL-12, type-I-interferon, IL-6, and direct JAK/TYK activity.
- Week-16 efficacy at least comparable with approved icotrokinra (proposed minimum PASI 90 ≥55%; target ≥60%), plus a credible second advantage such as food-independent exposure or simpler manufacturing.
- A structurally verified binding mode and clean artifact/cytokine counterscreens before large chemistry investment.

Evidence hierarchy and scope

Table 2. Validation hierarchy

Tier	Evidence	Examples	Use in this report
A	Regulator-reviewed human evidence	FDA labels, reviews, randomized pivotal trials	Product benchmarks and safety context
B	Peer-reviewed human / translational evidence	Genetics, lesional tissue, human cells, clinical pharmacology	Target validation and biomarker rationale
C	Experimental structural evidence	X-ray and cryo-EM coordinates; primary structural papers	Binding-site and construct-aware design
D	Reproduced preclinical pharmacology	Orthogonal biochemical, cellular, and model data	Candidate gates
E	Exploratory or sponsor-only	Small cohorts, virtual screening, docking/MD without direct binding	Hypothesis generation only

Definition used here. “Small molecule” means a chemically defined, non-peptide oral agent, ideally Rule-of-Five-like but allowing justified beyond-Rule-of-Five space. Icotrokinra and other oral cyclic peptides validate the extracellular mechanism but are comparators, not proof that a conventional small molecule is feasible.

1. Target biology and validation

1.1 Mechanistic proposition

IL-23 is a disulfide-linked heterodimer of the unique p19 subunit (IL23A) and the p40 subunit (IL12B), which is shared with IL-12. IL-23 first engages the D1 domain of IL23R through a p19-dominant site-3 interface; this primes recruitment of IL12Rβ1 through p40 and enables TYK2/JAK2-mediated signaling, predominantly through STAT3. Downstream maintenance of pathogenic type-17 immunity supports IL-17A/F, IL-22, CCL20, and keratinocyte inflammatory programs. [1,2,6,7]

A direct p19-IL23R disruptor is upstream and extracellular. In contrast with TYK2 inhibitors, it should selectively interrupt IL-23 while leaving IL-12 signaling through p40/IL12Rβ1 intact. That selectivity is a design hypothesis—not an entitlement—and must be demonstrated in IL-12→pSTAT4/IFN-γ and broader cytokine counterscreens.

Direct IL-23 receptor blockade preserves IL-12 signaling

The proposed molecule binds extracellular p19 and prevents IL23R recruitment; it is not a TYK2/JAK inhibitor.

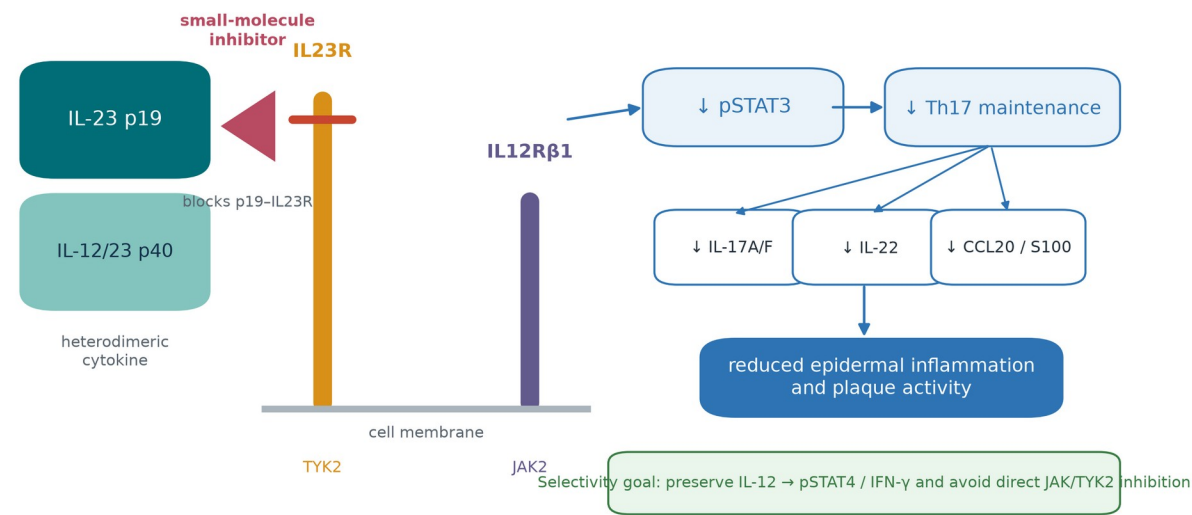


Figure 1. Mechanism of a direct p19-IL23R inhibitor and intended pathway selectivity.

Schematic derived from receptor-assembly structures and IL-23 pharmacology [2,6,7,13].

1.2 Convergent human validation

Table 3. Target-validation evidence stack

Evidence axis	Observation	Drug-discovery interpretation	Confidence
Lesional expression	IL23A/p19 and IL12B/p40 transcripts were reported ~22-fold and ~12-fold higher in psoriatic lesions than non-lesional skin, whereas IL12A/p35 was not similarly elevated.	Supports the IL-23 rather than IL-12 arm; expression alone is not causal.	High [3]
Common genetics	IL23R and IL12B variants associate with psoriasis; IL23R R381Q is protective.	Human lifelong pathway perturbation supports causal relevance, but R381Q is intracellular and does not validate the extracellular pocket.	High [4,5]

Evidence axis	Observation	Drug-discovery interpretation	Confidence
Functional genetics	Gln381 attenuates IL-23 responsiveness, pSTAT3, IL-17A/F, and memory-Th17 responses.	Supports partial IL23R inhibition and proximal pSTAT3 as a biomarker.	High [5]
Biologics	Guselkumab, tildrakizumab, and risankizumab show reproducible, high psoriasis efficacy by selectively neutralizing p19.	Strong clinical validation of p19; parenteral durability defines the efficacy ceiling.	Very high [16–18]
Direct oral pharmacology	Icotrokinra is an approved 13-aa oral macrocycle that competitively binds IL23R and selectively blocks IL-23.	Definitive human validation of receptor-side extracellular PPI antagonism; not conventional-small-molecule validation.	Very high [13–15]
Human loss of function	Complete inherited IL23R deficiency causes susceptibility to mycobacterial disease and sometimes mucocutaneous candidiasis.	Frames on-target host-defense risk; complete lifelong deficiency is not equivalent to partial, reversible dosing.	High [20]

1.3 Pharmacology that anchors preclinical translation

Icotrokinra (JNJ-77242113/PN-235) is the most informative comparator because its mechanism is directly aligned. In peer-reviewed assays it bound human IL23R ectodomain with $KD\ 7.1 \pm 2.5\ \mu M$, inhibited IL-23-induced pSTAT3 in human PBMCs with $IC_{50}\ 5.6 \pm 1.2\ \mu M$, and showed no meaningful IL-12 pSTAT4 inhibition at concentrations above $2\ \mu M$. Median whole-blood IL-23 $\rightarrow$ IFN- γ IC_{50} values were 11 pM in healthy donors and 9 pM in psoriasis samples. Oral dosing inhibited the pathway in humans and suppressed IL-23-driven skin thickening and IL-17A/F/IL-22 expression in rats. [13]

Translation rule Benchmark a new compound against icotrokinra at matched pathway inhibition, not matched nominal concentration. A conventional molecule may be less intrinsically potent yet viable if free oral exposure, tissue penetration, food independence, and manufacturability compensate.

1.4 On-target safety hypothesis

- Expected class risks are infections—particularly TB/NTM and fungal infection—plus live-vaccine restrictions; current direct and antibody labels support baseline infection/TB screening. [14,16–18,20]
- The desired small molecule should not add kinase-related cytopenia, lipid, hepatic, muscle, cardiovascular, or broad immunosuppression liabilities.
- Preclinical host-defense experiments should compare efficacy-matched partial and near-complete blockade with icotrokinra and anti-p19 controls; greater suppression than those benchmarks is a kill signal.

2. Clinical and competitive landscape

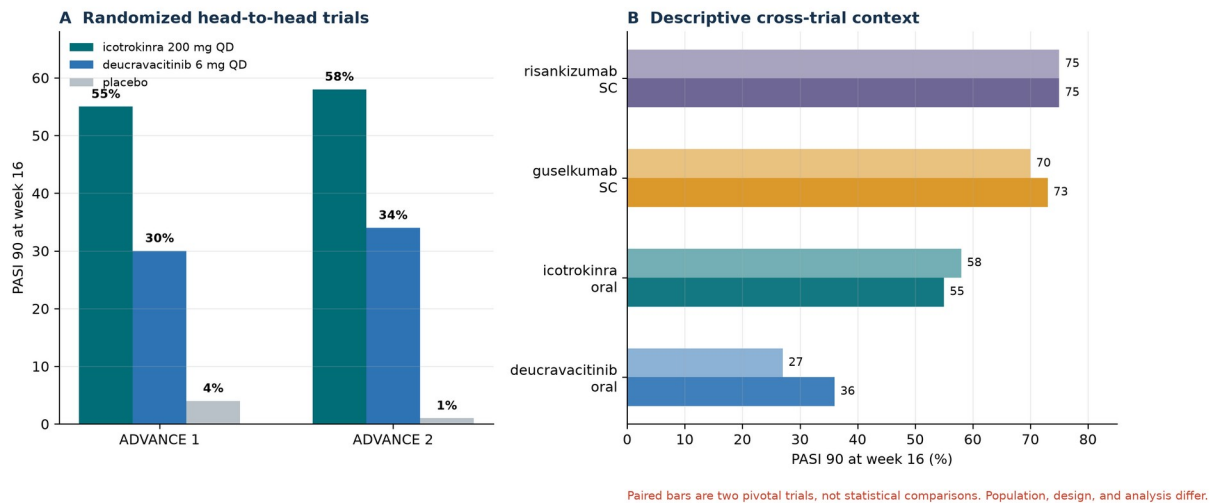
The competitive conclusion is unambiguous: oral direct IL23R blockade is no longer a modality concept—it is an approved psoriasis mechanism. The remaining white space is for a genuinely small, non-peptide molecule with a lower dose, less food sensitivity, or another meaningful product advantage. No public clinical-stage conventional low-molecular-weight direct IL-23/IL23R PPI inhibitor was identified at the cutoff; confidence is moderate because undisclosed programs may exist.

Table 4. Competitive landscape at 1 September 2026

Asset	Mechanism / modality	Status	Key evidence	Discovery implication
Guselkumab	SC anti-p19 mAb	Approved	W16 PASI90 73%/70%; IGA 0/1 85%/84% in VOYAGE-1/2	High-efficacy, durable benchmark [16]
Tildrakizumab	SC anti-p19 mAb	Approved	W12 PASI75 64%/61% in reSURFACE-1/2	Cross-antibody validation; q12w convenience [17]
Risankizumab	SC anti-p19 mAb	Approved	W16 PASI90 75%/75%; PASI100 36%/51%	Efficacy/durability ceiling [18]
Icotrokinra / ICOTYDE	13-aa oral macrocyclic IL23R antagonist	FDA-approved 17 Mar 2026	ADVANCE-1/2 W16 PASI90 55%/58% vs deucravacitinib 30%/34% and placebo 4%/1%	Direct oral mechanism proven; 200 mg QD fasting and food effect leave differentiation room [14,15]
AVR-001 / HS-20118	Oral cyclic IL23R antagonist	Early clinical; Phase 2b planned	Small sponsor-reported psoriasis cohort; no regulator-reviewed efficacy	Potential long-acting oral peptide; evidence low confidence [22,23]
PTG-200; PN-232	Earlier oral IL23R peptides	Discontinued Q4 2021	Superseded by PN-235 for potency/PK/PD	Shows optimization burden and attrition [24]
Peptide 23-446	p19-binding cyclic peptide	Preclinical tool	PDB 8UUI; optimized analogues low-nM; FP assay and fragment pilot	Best public induced-pocket precedent [10,29]
Deucravacitinib	Oral allosteric TYK2 inhibitor	Approved	POETIK W16 PASI90 36%/27%; PASI75 58%/53%	Approved oral efficacy/safety benchmark; broader cytokine mechanism [19]
Zasocitinib; envudeucitinib	Next-generation oral TYK2 inhibitors	Phase 3; not approved at cutoff	Sponsor-only W16 PASI90 ranges ~52–61%	Crowded oral market; regulator review pending
Inh-31	Docking-derived low-MW candidate	Exploratory preclinical	HEK-Blue IL-23 SEAP IC50 22.6 μ M; docking/MD; no convincing direct biophysical engagement or in-vivo efficacy	Hypothesis only; not a validated chemical starting point [21]

Regulator-reviewed values are from current FDA labels/snapshots; sponsor-only disclosures are explicitly labeled [14–19,22–24].

Clinical proof is strong; the differentiation bar for another oral therapy is high



Source: FDA labels/trial snapshots. Efficacy values are rounded label-reported proportions.

Figure 2. Psoriasis efficacy benchmarks for approved direct IL23R, TYK2, and anti-p19 therapies.

Panel A is randomized head-to-head evidence. Panel B is descriptive cross-trial context only; study populations and analyses differ [14–19].

2.1 Product differentiation implications

- Mechanism alone is not differentiation. Direct oral IL23R antagonism is already approved.
- Icotrokinra is 200 mg once daily on an empty stomach; a high-fat meal lowers AUC by 43% and C_{max} by 59%. A lower-dose, food-independent tablet is a concrete opportunity. [14]
- PASI 90 around 55–60% is now the minimum competitive oral range; anti-p19 biologics remain around 70–75% in pivotal trials. [14,16,18,19]
- Safety positioning must be comparator-based. Icotrokinra's label includes infection/TB/live-vaccine warnings and observed fungal infection; a new molecule should not add compound-specific monitoring burdens. [14]
- A portfolio should demand at least two credible advantages among dose, food independence, efficacy, safety/monitoring, cost of goods, or convenience before candidate nomination.

3. Structural biology and ligandability

The p19-IL23R site is structurally well characterized but intrinsically difficult. The reported major interface is approximately 900 Å² and involves a distributed contact network rather than a single deep cavity. IL23R D1 supplies an N-terminal hook and a hydrophobic/aromatic cradle around a W156/L160-centered p19 region. A smaller IL23R-p40 contact also exists, so “IL23R binds only p19” is an oversimplification. [6–9]

Table 5. Experimental structural atlas

PDB	Method	Resolution	Assembly	Use and caveat
5MXA	X-ray	2.50 Å	Apo IL-23 p40/p19	Apo comparator for receptor- and ligand-induced rearrangements [25]
5MZV	X-ray	2.80 Å	IL-23 + IL23R ectodomain + remote p40 nanobody	Highest-value atomic map of site 3; soluble/glycoengineered construct [6,26]
6WDQ	X-ray	3.40 Å	Partial quaternary IL-23/IL23R/IL12Rβ1 assembly	Confirms contact pattern and receptor-orientation plasticity; IL23R L310P and truncated IL12Rβ1 [7,27]
8OE4	cryo-EM	3.60 Å	Complete extracellular assembly	Global architecture/accessibility; engineered CaM-DAPK predimerization and p40 N303D [9,28]
8UUI	X-ray	2.43 Å	IL-23 + cyclic peptide 23-446	Best ligandability model: cyclic-peptide-induced cryptic p19 pocket [10,29]
8CR8	X-ray	2.00 Å	Additional apo IL-23 ensemble	High-resolution apo-state comparison; p40 N303D
6UIB	X-ray	2.74 Å	IL-23 + distal-binding peptide	Useful distal-site counterscreen structure
5MJ3 / 3D85	X-ray	1.74 / 1.90 Å	p19/p40 with protein binders	Additional conformational ensemble; not direct small-molecule templates

Resolutions and constructs from RCSB entries and primary structural publications [6,7,9,10,25–29].

Structural map: quaternary assembly, receptor interface, and inducible p19 pocket

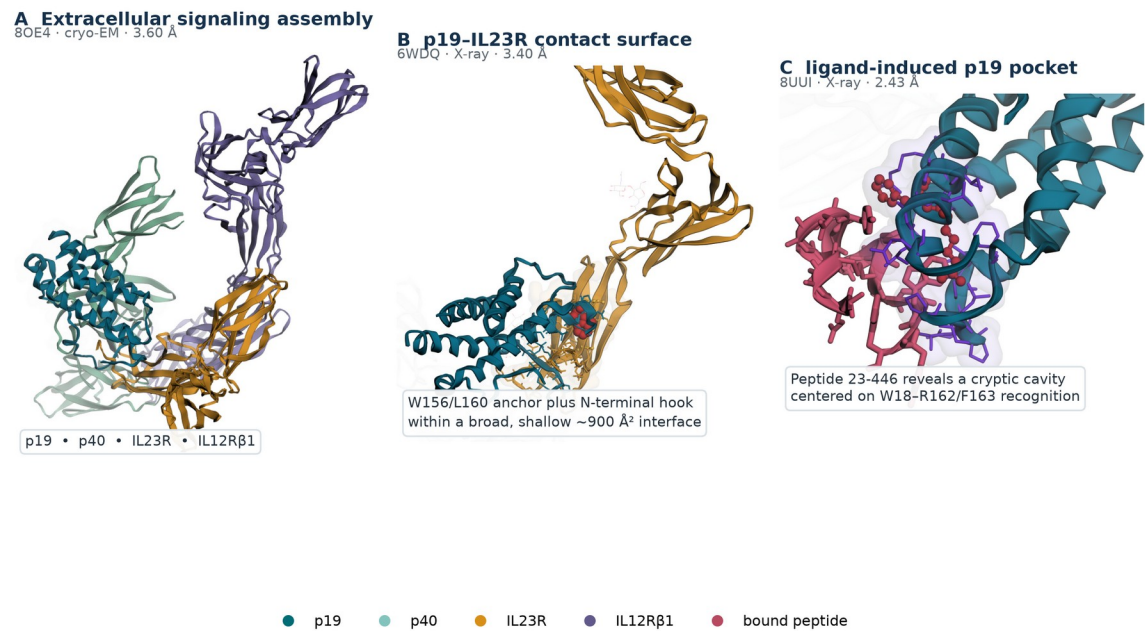


Figure 3. Structural map of the extracellular receptor assembly, the p19-IL23R interface, and the induced p19 pocket. Coordinate-based renderings of PDB 8OE4, 6WDQ, and 8UUI. Colors are consistent across panels [9,10,27–29].

3.1 Interface anatomy

Table 6. Reported site-3 contact clusters

Cluster	p19 side	IL23R side	Interpretation
N-terminal hook / AB-loop clamp	L56, R57, E58, E59, G60, K164	G24–N29	K164 contacts hook backbone; receptor binding orders the p19 AB loop.
W156/L160-centered anchor	Q154, P155, W156, Q157, L160, L161	I28, Y100, L113, G116, D118	W156 stacks near G116 and its backbone contacts D118; combined W156A/L160E is much more disruptive than W156 alone.
Hydrophobic/polar shelf	L159, F163, I165, R167, S168, L169	M99, T102, H108, F109, Q110, E111, T112, C115, K117	Coordinate contacts define vectors; most residues are not experimentally proven energetic hotspots.
Secondary p40 contact	p40 D109 region	IL23R R62 region	Small ~260 Å² contact; may influence orientation but is not the main design site.

Residues use human full-precursor/UniProt numbering; coordinate proximity is not synonymous with energetic hotspot status [6–8,26,27].

3.2 The 8UUI inducible pocket

Peptide 23-446 remodels p19 into a cavity absent or collapsed in apo structures. The modeled peptide residues place W18 deepest in a pocket involving p19 Pro113/Ser114/Leu115/Leu116/Ser119/Pro120/Val121/Leu159/Leu160/Arg162/Phe163/Leu166. W18 engages the Arg162 region; Phe14/Tyr10 address Phe163; and polar contacts project toward the

Ser114/Leu116 backbone. Optimized peptides reached KD values of ~16–20 nM and PPI IC50 values of ~28–64 nM. [10,29]

The associated 12,800-compound pilot fragment screen was technically credible (reported Z' ≈0.70), but confirmed direct binders remained weak (KD ~180 μM to 1.95 mM). No fragment co-structure, cellular target engagement, or demonstrated fragment-mediated receptor blockade was reported. Therefore 8UUI is evidence for a ligandable conformational state—not a drug-like lead. [10]

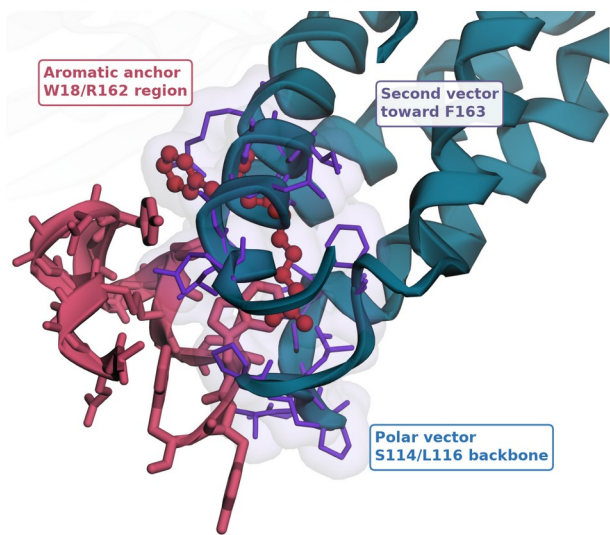
3.3 Structural cautions for design

- Treat the target as an ensemble. The receptor orientation changes by roughly 18° between reported assemblies, and construct/crystal effects may contribute. [6,7]
- Use 5MZV/6WDQ for atomic interface design, 8OE4 for global architecture only, and 8UUI for pocket-based docking and fragment growth.
- Do not overstate W156 as a single human Achilles heel: W156 is central structurally, but human signaling is distributed and combined W156/L160 mutation is more decisive. [8]
- Use mammalian, glycosylated human proteins. IL23R D1 glycans (including N47/N81) may alter access, although their precise pocket effect is unproven.
- There is no obvious validated solvent-exposed nonstructural cysteine in current site-3 structures; the IL23R C30–C115 disulfide is structural, so covalent design is not a lead hypothesis.

4. Small-molecule discovery strategies

Four structural strategies—only one starts with direct ligandability evidence

Priority route: 8UUI inducible p19 pocket



Scores integrate structural evidence, chemical tractability, assayability, selectivity, and translational validation.



Caution: 8UUI validates a pocket, not a drug-like lead; reported fragment KDs remain ~0.18-1.95 mM.

Integrated feasibility score (5 = strongest)

Figure 4. Ranked structural strategies for direct p19-IL23R inhibition.

Scores integrate public structural evidence, chemical tractability, assayability, selectivity, and translational validation [6–13].

Table 7. Strategy scorecard (1 = poor; 5 = strong)

Rank	Strategy	Structural evidence	SM tractability	Mechanistic confidence	Priority	Principal red flag
1	Grow/link fragments in 8UUI pocket	5	3	3	4.0	Pocket was induced by a large peptide; weak fragments may bind without blocking IL23R.
2	Conformational lock of p19	3	3	2	3.0	No small molecule has yet shown that locking the observed state prevents signaling.
3	W156/L160-centered mimetic	4	2	4	2.0	Shallow receptor cradle likely requires rigid 3D/bRo5 or macrocyclic chemistry.
4	Direct interface-spanning ligand	5	1	4	1.0	Broad distributed interface creates large desolvation and entropy penalties.

4.1 Priority: fragment growth in the inducible p19 pocket

- Anchor one aromatic/heteroaromatic element in the W18/Arg162 subpocket, preferably with tunable cation- π and desolvation balance.
- Project a second aromatic vector toward Phe163 while avoiding peptide-like planar surface area and excessive lipophilicity.
- Use a directional donor/acceptor vector toward the Ser114/Leu116 backbone region; constrain geometry with fused or spirocyclic 3D scaffolds.
- Grow into the receptor-overlap volume and prove PPI displacement. Binding deeper in the cavity without functional obstruction is insufficient.
- Screen apo, peptide-bound, and receptor-bound ensembles. Favor crystallographic fragment soaking/co-crystallization, ligand-observed NMR, SPR, and high-quality FP/TR-FRET over rigid single-state docking.

Hit-generation mix. Use a crystallography-enabled fragment library, focused virtual/shape screening seeded by peptide pharmacophore vectors, and a diversity HTS against the validated FP/displacement assay. A DNA-encoded library can be considered only with strong solution-phase confirmation because immobilization and avidity can exaggerate extracellular PPI hits.

4.2 Backup: ligand-induced conformational lock

Receptor and peptide binding remodel the p19 AB loop, helix-D region, and the 110–120/159–166 network. A ligand could inhibit by stabilizing a receptor-incompetent conformation even if it only partially overlaps the orthosteric footprint. This route is attractive for smaller chemotypes but mechanistically risky: every series needs structural confirmation, receptor-displacement data, and mutation-rescue or state-selective biophysics to separate causal conformational locking from nonspecific binding.

4.3 Higher-molecular-weight backup: mimic the W156/L160 region

The receptor-side cradle suggests a rigid, three-dimensional scaffold that mimics the W156 aromatic anchor and L160 hydrophobe while reproducing the D118 backbone-contact vector and engaging the N-terminal hook. An indole/azaindole-like core, bicyclic peptidomimetic, constrained bRo5 scaffold, or macrocycle is more plausible than a conventional flat fragment. Receptor-binding macrocycles and icotrokinra validate this surface, but published receptor poses are not crystallographic proof of a small-molecule pocket. [11–13]

4.4 Deprioritize: spanning the native flat interface

A ligand that directly spans the native p19–IL23R surface would face the highest entropy, desolvation, and size burden. This route should not drive broad screening unless new structural data reveal a persistent subpocket or a compact hotspot combination. Inh-31 does not change that conclusion: its 22.6 μ M cell-reporter activity and docking/MD analysis lack sufficiently convincing direct target engagement to establish the binding mode. [21]

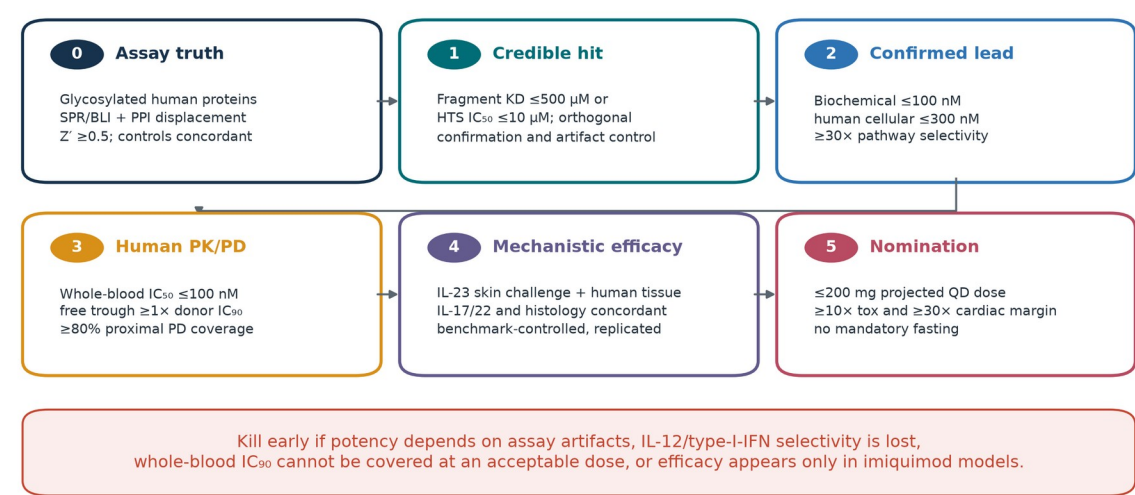
First structural kill criterion Do not progress a series on pocket affinity alone. Require a reproducible concentration-dependent relationship among direct binding, p19–IL23R displacement, full-length cell signaling, and an on-target structural or mutational signature.

5. Preclinical discovery and translation plan

The campaign should be organized around falsifiable links: chemical identity → direct binding → PPI displacement → native-cell pathway inhibition → oral free-exposure coverage → human-relevant skin pharmacology. Assay concordance and matched comparators matter more than any single potency number.

A gated discovery cascade connects direct binding to human skin pharmacology

Advance only when orthogonal assays, selectivity, exposure, and mechanistic biomarkers agree.



Reference comparators throughout: icotrokinra (receptor-side pharmacology), anti-p19 antibody, inactive analogue, and a selective TYK2 inhibitor.

Figure 5. Gated discovery cascade from assay qualification to candidate nomination.

Thresholds are proposed program gates, not observed properties of an existing compound. IC_{90} assumptions require measured Hill slopes.

5.1 Assay architecture and acceptance criteria

Table 8. Assay and mechanism-validation package

Module	Recommended implementation	Decision purpose
Protein quality	Mammalian, glycosylated human p19/p40 and IL23R D1/ECD; monodispersity, activity, stoichiometry	Preserves native disulfides/glycans and avoids false construct-specific hits
Direct binding	SPR/BLI or MST plus ligand-observed NMR/DSF where appropriate	Report kinetics and equilibrium; require concentration dependence and stoichiometry
PPI displacement	FP based on validated peptide probe; orthogonal TR-FRET/AlphaLISA or SPR competition	$Z' \geq 0.5$; positive-control IC_{50} within threefold; dilution and detergent controls
Mechanism controls	Icotrokinra, anti-p19, peptide 23-446, inactive analogue, distal-site peptide/structure	Separates p19-pocket blockade, receptor-side blockade, distal binding, and assay interference
Artifact counterscreens	Aggregation, fluorescence, redox/reactivity, nonspecific protein binding, cytotoxicity	Reject PAINS-like behavior and colloidal/nuisance mechanisms
Mutational mapping	Pocket residues (e.g., S114/L116/R162/F163) and interface controls with folding checks	A potency shift supports—but does not alone prove—the binding mode
Native-cell signaling	Full-length IL23R/IL12R β 1 cells; pSTAT3 at 15–30 min; distal IFN- γ /IL-17/IL-22	Links direct binding to signaling and avoids reporter-only artifacts

Module	Recommended implementation	Decision purpose
Pathway selectivity	IL-12→pSTAT4/IFN-γ; IFNα→pSTAT1; IL-6→pSTAT3; direct JAK/TYK panel	Minimum ≥30-fold; target ≥100-fold functional separation

5.2 Human cellular and tissue cascade

1. PBMC and engineered-cell pSTAT3: establish proximal potency, kinetics, and receptor dependence.
2. Healthy and psoriasis whole blood: measure donor distribution for IL-23-stimulated pSTAT3 and/or IFN-γ; use the 90th-percentile donor IC90 for coverage projections.
3. Patient DC/Th17/keratinocyte co-culture: quantify IL-17A/F, IL-22, keratinocyte activation, and viability at matched exposure.
4. Lesional explant or psoriasis-skin xenograft: require normalization of a prespecified type-17/keratinocyte signature across at least three independent donors.

5.3 Biomarker framework

Table 9. Translational biomarker hierarchy

Layer	Readouts	Matrix / timing	Purpose
Target engagement	Direct ligand displacement; receptor occupancy or accessible-site occupancy; competition with labeled probe	Biochemical / whole blood	Confirm exposure reaches the intended extracellular target
Proximal PD	IL-23→pSTAT3 in NK/MAIT/memory T cells; ex-vivo stimulated whole-blood IFN-γ	15–30 min / hours	Primary concentration–effect and coverage anchor
Selectivity	IL-12→pSTAT4/IFN-γ; IFNα→pSTAT1; IL-6→pSTAT3; IL-2/15 and GM-CSF panels	Matched timepoints	Demonstrate p19–IL23R rather than kinase blockade
Distal immune	IL-17A, IL-17F, IL-22, IFN-γ	24–72 h	Confirm functional pathway consequences
Skin molecular	IL17A/F, IL22, DEFB4A, S100A7/A8/A9, LCN2, IL36G, CCL20, KRT16/17	Biopsy/explant	Connect pathway PD to psoriasis-relevant tissue response
Skin cellular / histology	K16, Ki67, epidermal thickness, CD3, CD11c	Biopsy/model	Orthogonal tissue-level efficacy
Clinical translation	PASI, IGA, BSA, itch/PSSD	Future clinical	Link biomarker depth and exposure to efficacy

5.4 Model hierarchy

Table 10. Preclinical model hierarchy

Model	Strength	Limitation	Recommended use
IL-23 intradermal skin challenge	Mechanism-proximal, exposure/PD link, IL-17/22 and hyperplasia	Acute and species/cross-reactivity dependent	Primary in-vivo PoC; benchmark against anti-p19 and icotrokinra [13,31]
Human IL23R knock-in / humanized model	Resolves cross-species potency and receptor pharmacology	Construct fidelity and immune context	Use if candidate lacks relevant-species cross-reactivity
Patient skin explant / organotypic co-culture	Human lesion biology and molecular signature	Viability, donor variability, limited systemic PK	Required human-tissue bridge before nomination

Model	Strength	Limitation	Recommended use
Psoriasis skin xenograft	Human tissue architecture and repeated dosing	Complexity, throughput, graft variability	Optional high-value confirmation
Imiquimod dermatitis	Cheap, robust supportive inflammation model	Acute innate/ $\gamma\delta$ -T-cell dependence; strain/sex effects; limited disease specificity	Supportive only; efficacy here cannot rescue failure in mechanistic/human systems [30]

5.5 PK/PD and species strategy

- Measure unbound plasma and skin exposure, not total concentration alone; connect both to ex-vivo whole-blood PD.
- Minimum projection: human free trough at least 1× the 90th-percentile donor whole-blood IC90; target 3× throughout the interval.
- Do not assume IC90 from IC50. With a unit Hill slope IC90 is ~9× IC50; measure slopes in relevant matrices.
- Use an oral cross-reactive species when possible. If receptor pharmacology is human-specific, combine a humanized model with standard nonclinical safety species and human-cell off-target work.
- Require ≥80% proximal PD inhibition across the intended interval at the in-vivo efficacy exposure; confirm exposure response and reversal on washout.

6. Preclinical target product profile

Status of values. Every number in Tables 11–12 is a proposed program target. Clinical efficacy, durability, monitoring, and shelf-life rows describe the intended future product; they cannot be proven at the preclinical stage. Comparator data are cited separately.

Table 11. Future clinical-use and product TPP

Attribute	Minimum viable profile	Target profile	Rationale / benchmark
Population	Adults with moderate-to-severe plaque psoriasis eligible for systemic therapy / phototherapy	Biologic-naïve and experienced adults; adolescent expansion after adult proof	Approved pathway population [14–19]
Mechanism	Direct reversible p19–IL23R blockade; preserve IL-12	Defined orthosteric/allosteric binding mode; no meaningful IL-12, type-I-IFN or direct JAK/TYK inhibition	Mechanistic differentiation
Modality	Non-peptide, chemically defined oral molecule; MW preferably ≤750 Da if bRo5 is justified	MW ≤550 Da with Rule-of-Five-like properties	Aspirational chemistry preference
Administration	≤200 mg once-daily tablet; no enhancer; no mandatory fasting	≤50–100 mg QD; fed/fasted AUC and Cmax ratios 0.8–1.25	Must improve on 200 mg fasting icotrokinra [14]
Week-16 efficacy	PASI90 ≥55%; IGA 0/1 ≥68%; PASI75 ≥75%	PASI90 ≥60%; IGA 0/1 ≥72%; PASI75 ≥80%; PASI100 ≥30%	At least approved direct-oral range [14,15]
Onset	Clinically meaningful separation by week 4–6	Clear itch and skin response by week 4	Clinical aspiration
Durability	≥75% of week-24 PASI90 responders retain response at week 52	≥85% retain response	Clinical aspiration; biologic durability benchmark [18]
Clinical safety	No worse than approved selective oral benchmark; manageable infection risk	No compound-specific laboratory monitoring; no added kinase-like liabilities	Must be established clinically
Monitoring	Baseline infection/TB assessment and vaccination review acceptable	No routine laboratory monitoring after baseline	Clinical aspiration
Meaningful differentiation	At least two advantages vs icotrokinra among dose, food effect, efficacy, safety, cost, convenience	Low-dose food-independent QD tablet plus one further advantage	Portfolio continuation rule

Table 12. Preclinical candidate profile

Candidate attribute	Minimum gate	Target	Qualification
Biochemical potency	KD or competition IC50 ≤30 nM	≤10 nM	Candidate-quality binding
Human whole-blood potency	IL-23 IC50 ≤100 nM; IC90 ≤1 μM	IC50 ≤30 nM; IC90 ≤300 nM	Across ≥10 healthy and ≥6 psoriasis donors
Functional selectivity	≥30× over IL-12, IFNα, IL-6 and direct kinase effects	≥100× and clean broader panel	Prevents TYK2/JAK-like mechanism
PK/PD coverage	Projected human free trough ≥1× donor-90th-percentile whole-blood IC90	≥3× IC90 throughout interval	Coverage, not nominal potency, drives dose
Oral PK	F ≥15% or sufficient exposure for ≤200 mg QD; effective coverage compatible with QD	F ≥30%; t½ ≥12 h or equivalent slow-offset coverage	Preclinical projection
DDI	No strong/time-dependent CYP inhibition/induction or critical transporter liability	Off-target/DDI IC50 ≥50× free Cmax	Avoid chronic-treatment restrictions

Candidate attribute	Minimum gate	Target	Qualification
General safety	Genotox negative; no material CV/respiratory/CNS safety-pharmacology signal	Clean chronic tox without compound-specific monitoring	Candidate gate / clinical aspiration
Tox margin	$\geq 10\times$ unbound human exposure to NOAEL/findings	$\geq 30\times$	Minimum / target
Cardiac margin	hERG/multichannel $\geq 30\times$ free Cmax plus clean in-vivo ECG	$\geq 100\times$	Minimum / aspirational target
Host defense	No greater mycobacterial/Candida suppression than icotrokinra or anti-p19 at matched efficacy	Efficacy with measurable host-defense reserve	On-target risk management
Formulation	Stable immediate-release tablet suitable for clinical supply	Small tablet; room-temperature stability ≥ 24 months	Shelf life is aspirational in preclinical development
Projected dose	≤ 200 mg QD, no mandatory fasting	≤ 50 – 100 mg QD, food independent	Hard portfolio gate / target

TPP discipline PASI90 $\geq 70\%$, dose ≤ 50 mg, $\geq 100\times$ cytokine selectivity, $\geq 30\times$ tox margin, $\geq 100\times$ cardiac margin, no routine monitoring, and 24-month shelf life are useful aspirations—not preclinical minimums. The minimum profile is deliberately set at a level that can support a rational development decision.

7. Risks, stage gates, and recommendation

7.1 Stage-gate package

Table 13. Proposed stage gates

Gate	Advance criterion	Kill / recycle trigger
0 — Assay qualification	Z' ≥ 0.5 ; positive control within threefold; concordant direct binding + displacement/cell assay; expected mutant behavior	Assay artifacts or construct-dependent pharmacology cannot be resolved
1 — Hit confirmation	Fragment KD ≤ 500 μ M with LE ≥ 0.25 kcal/mol/heavy atom and structure/orthogonal confirmation; or HTS IC50 ≤ 10 μ M	No genuine binders after two independent, well-powered campaigns
2 — Confirmed lead	Biochemical ≤ 100 nM; human cellular ≤ 300 nM; $\geq 30\times$ pathway selectivity; clean aggregation/interference	Potency requires nuisance chemistry or loses IL-12/type-I-IFN selectivity
3 — Optimized lead	Whole-blood IC50 ≤ 100 nM, IC90 ≤ 1 μ M across donor panel; oral exposure covers free IC90	Projected dose >200 mg QD or required BID/fasting regimen
4 — Mechanistic in-vivo PoC	Replicated blinded IL-23 skin studies; exposure response; $\geq 50\%$ reversal of disease-control-excess thickening and $\geq 70\%$ IL-17/22 composite suppression	Target engagement without expected pathway/tissue response
5 — Human tissue PoC	$\geq 50\%$ prespecified signature normalization in ≥ 3 psoriasis donors, within 20% of anti-p19 reference, $\leq 10\%$ viability loss	Fails both human tissue and mechanistic IL-23 model despite full exposure/engagement
6 — Candidate nomination	≤ 200 mg projected QD; no mandatory fasting; genotox negative; $\geq 10\times$ tox and $\geq 30\times$ cardiac margin; viable species plan	Profile lacks two advantages vs icotrokinra or has non-mitigable tox/DDI
7 — IND risk closure	Chronic tox, safety pharmacology, genotox, DDI, reproductive plan, host-defense/TB/vaccine plan	Opportunity or infection liability produces unacceptable benefit-risk

7.2 Risk register

Table 14. Preclinical risk register

Risk	Likelihood	Impact	Mitigation	Kill criterion
PPI not ligandable by a conventional molecule	High	High	Parallel cryptic-pocket and receptor-hotspot tracks; fragment crystallography/NMR/SPR	No structurally validated series ≤ 100 nM after two independent campaigns
Potency-exposure mismatch	High	High	Optimize free exposure, solubility, clearance and residence time together; early human dose projection	>200 mg QD, BID required, or trough cannot cover IC90
Food effect / variable absorption	Med-high	High	Biorelevant dissolution, permeability, early fed/fasted studies, formulation	Unfixable ≥ 2 -fold food effect or mandatory prolonged fasting
Models mispredict human skin	High	High	IL-23 challenge + human tissue; IMQ supportive only	Works only in IMQ and fails both human tissue and mechanistic model
Complete engagement lacks competitive efficacy	Medium	High	Benchmark molecular signature and pathway suppression vs icotrokinra/anti-p19	Near-complete proximal PD gives $<50\%$ human-tissue normalization or $>30\%$ deficit vs reference
Loss of pathway selectivity	Medium	High	Early IL-12/IFN α /IL-6 and kinase counterscreens; genetic/structural mapping	$\leq 30\times$ separation at relevant free exposure

Risk	Likelihood	Impact	Mitigation	Kill criterion
TB/NTM/fungal host-defense risk	Medium	High	Matched-inhibition mycobacterial/Candida assays; dose-occupancy modeling; safety monitoring	Greater suppression than direct/anti-p19 comparators or serious opportunistic infection near efficacious exposure
Tox/DDI or competitive obsolescence	Med-high	High	Early reactive-metabolite, genotox, cardiac, CYP/transporter work; quarterly landscape/IP review	Genotox, cardiac <30×, tox <10×, or fewer than two meaningful product advantages

7.3 Recommended work packages

Table 15. Gated execution plan

Work package	Scope	Exit condition
A — Pocket truth	Qualify 8UUI-state assays; reproduce peptide pharmacology; fragment crystallography/SPR/NMR; demonstrate first ligand-mediated PPI displacement	A structurally credible fragment/hit with linked direct-binding and PPI effects
B — Parallel chemical hypotheses	Grow/link pocket hits; explore conformational lock; limited rigid W156/L160 mimetic set; develop SAR with matched property optimization	At least one series at ≤100 nM biochemical and ≤300 nM native-cell potency with ≥30× selectivity
C — Translation and nomination	Whole-blood donor panel; oral PK/PD; IL-23 skin challenge; human tissue; developability and safety package	Candidate meeting Table 12 minimums and at least two product advantages

7.4 Final recommendation

Proceed conditionally. The biology and therapeutic mechanism are unusually well validated for an extracellular PPI target. The conventional-small-molecule question is not. A disciplined campaign is justified because 8UUI supplies an inducible p19 pocket, low-nanomolar peptide pharmacology, a transferable assay, and weak but real fragment binding. The program should stop early if those fragments cannot be converted into structurally verified PPI inhibitors or if human whole-blood IC90 coverage projects beyond a practical oral dose.

Preferred portfolio shape. Lead with p19 pocket chemistry; keep a small, explicitly bRo5 receptor-hotspot effort as backup; avoid broad flat-interface screening. Nominate only a molecule that is selectively extracellular, orally coverable, no-fasting, and credibly differentiated from icotrokinra on at least two product attributes.

DECISION IN ONE LINE Strong target, credible new pocket, hard chemistry: GO only as a tightly gated ligandability-to-PPI-disruption experiment.

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Appendix A. Interpretation notes

- Affinity values from isolated ectodomains, membrane complexes, and cellular assays are not directly interchangeable; avidity, construct, glycosylation, and readout matter.
- Reported interface areas and contact counts are method- and structure-dependent, not invariant physical constants.
- Residues identified from coordinate proximity are contact-shell residues unless energetic importance is supported by mutagenesis or multiple structures.
- Clinical cross-trial values are context, not comparative-effect estimates. Only ADVANCE-1/2 provide randomized icotrokinra–deucravacitinib head-to-head evidence in this report.
- Sponsor-only data were included to map the competitive field and are explicitly downgraded in confidence.
- This is a drug-discovery planning document based on public information, not medical advice or a regulatory submission.